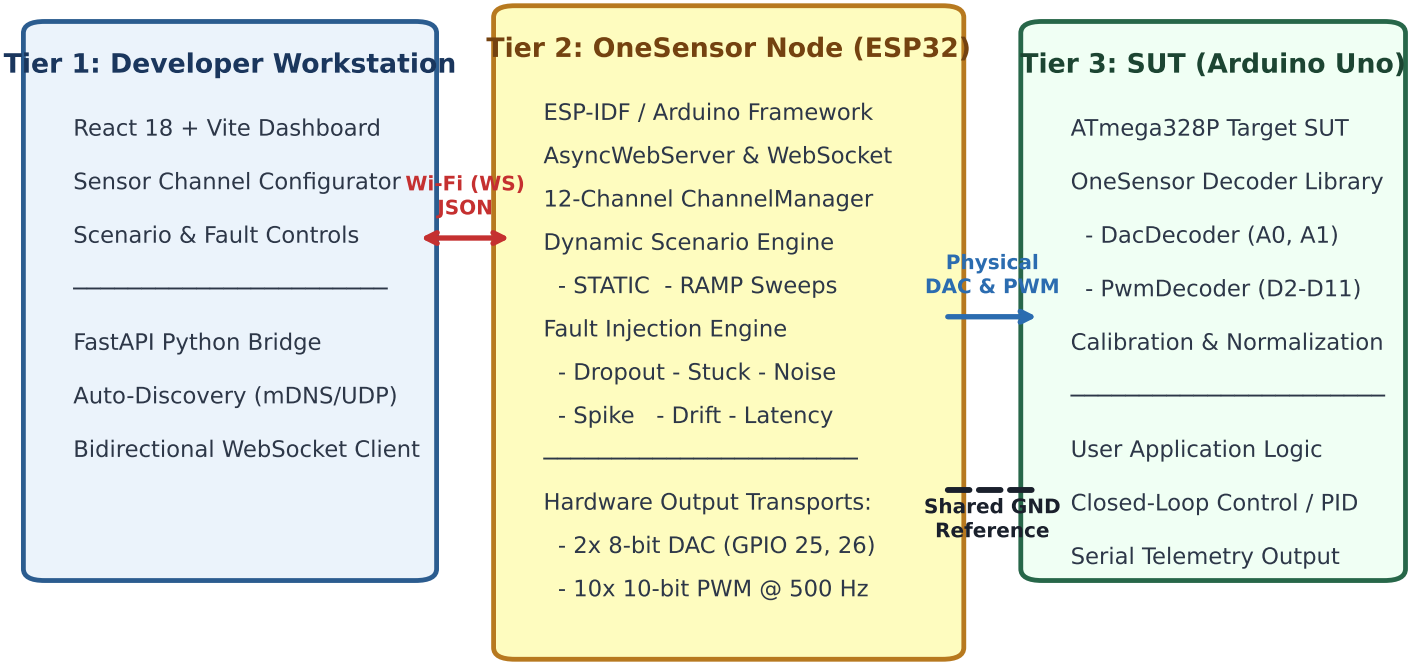
Tier 1: Developer Workstation
React 18 + Vite Dashboard
Sensor Channel Configurator
Scenario & Fault Controls
FastAPI Python Bridge
Auto-Discovery (mDNS/UDP)
Bidirectional WebSocket Client
Wi-Fi (WS)
JSON
Tier 2: OneSensor Node (ESP32)
ESP-IDF / Arduino Framework
AsyncWebServer & WebSocket
12-Channel ChannelManager
Dynamic Scenario Engine
- STATIC - RAMP Sweeps
Fault Injection Engine
- Dropout - Stuck - Noise
- Spike - Drift - Latency
Hardware Output Transports:
- 2x 8-bit DAC (GPIO 25, 26)
- 10x 10-bit PWM @ 500 Hz
Physical
DAC & PWM
Shared GND
Reference
Tier 3: SUT (Arduino Uno)
ATmega328P Target SUT
OneSensor Decoder Library
- DacDecoder (A0, A1)
- PwmDecoder (D2-D11)
Calibration & Normalization
User Application Logic
Closed-Loop Control / PID
Serial Telemetry Output